

Advanced (Habanero)

- Q1.** What is the $\bar{x}$ of the data set {2, 4, 6, 8, 10}?
- (A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q2.** A data set has a mode of 5 and a median of 7. What can be said about the data set?
- (A) It is symmetrical
(B) It is skewed to the left
(C) It is skewed to the right
(D) The mean is 6
(E) The mean is 5
- Q3.** What is the median of the data set {1, 3, 5, 7, 9, 11}?
- (A) 5
(B) 6
(C) 7
(D) 8
(E) 9
- Q4.** A data set has a mean of 10 and a median of 12. What can be said about the data set?
- (A) It is symmetrical
(B) It is skewed to the left
(C) It is skewed to the right
(D) The mode is 10
(E) The mode is 12
- Q5.** What is the mode of the data set {2, 4, 4, 6, 8}?
- (A) 2
(B) 4
(C) 6
(D) 8
(E) No mode
- Q6.** Two data sets have the same median. Does this mean they have the same mean?
- (A) Yes, always
(B) No, never
(C) Maybe, it depends on the data
(D) Only if the data is symmetrical
(E) Only if the data is skewed
- Q7.** A data set has a mean of 15 and a standard deviation of 3. What is the z-score of the data point 12?
- (A) -1
(B) -2
(C) -3
(D) 1
(E) 2
- Q8.** What is the $\bar{x}$ of the data set {10, 12, 14, 16, 18} if the data point 20 is added?
- (A) 14
(B) 15
(C) 16
(D) 17
(E) 18

Open Ended Questions (FRQ)

- Q9.** Explain the difference between the mean, median, and mode. Provide an example of when each would be the best measure of central tendency.
- Q10.** A bakery sells a variety of bread types. The prices of the bread are {2.50, 3.25, 3.50, 4.00, 4.50}. Calculate the mean, median, and mode of the prices. Which measure of central tendency best describes the data set?

Chef's Tips

- Emphasize the importance of understanding the context of the data when choosing a measure of central tendency.
- Remind students to check their calculations carefully, especially when working with decimals.
- Encourage students to use real-world examples to illustrate their understanding of the concepts.